Experiment – 2

Memory Recall Task UI Design & Chunking Analysis

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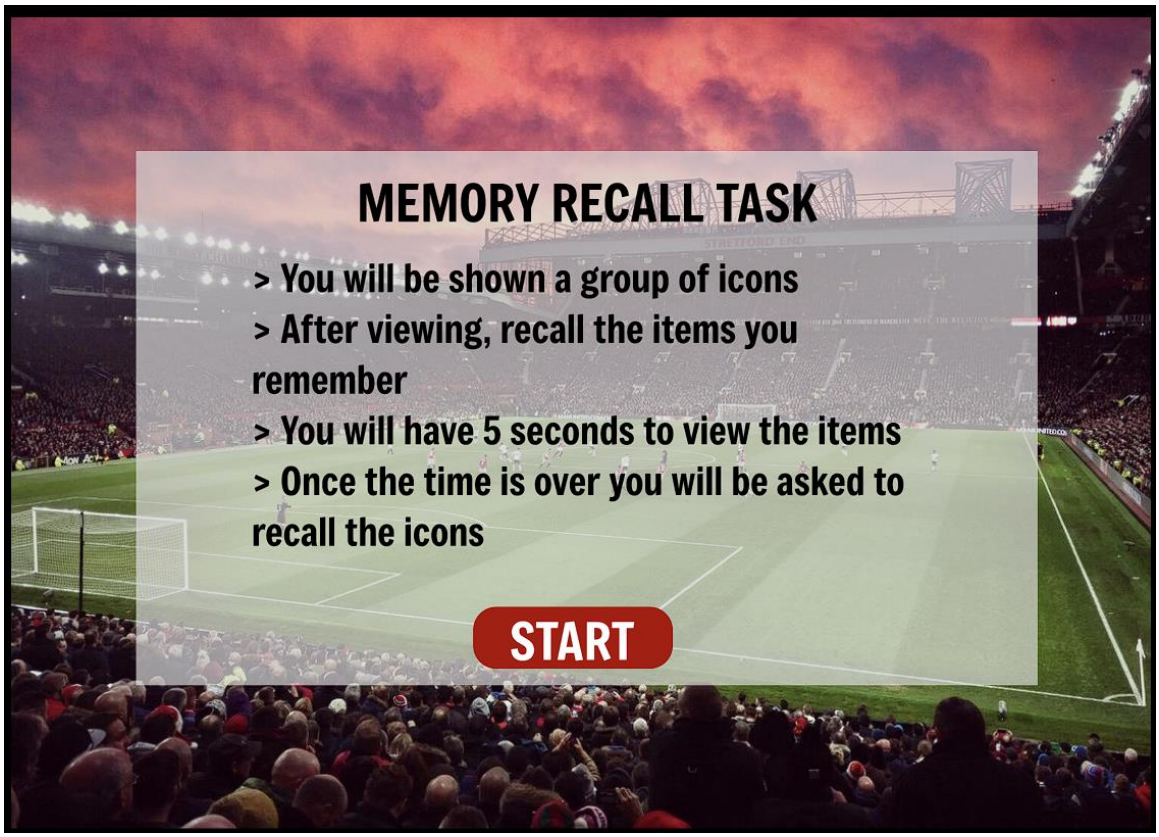
Aim of the Experiment:
To design and analyze a user interface that evaluates visual memory recall and to study how the chunking principle improves short-term memory performance.

FRAME 1: Instruction Page



This screen acts as the entry point of the memory recall task. Its primary goal is to prepare the user mentally for the upcoming activity by clearly explaining what they need to do.

Chunking is effectively applied by breaking instructions into short, numbered steps. Instead of presenting all information in a single paragraph, each instruction focuses on one action only, such as observing items, memorizing them, and recalling them later. This reduces cognitive overload and improves comprehension.

The large title "Memory Recall Task" establishes clarity, while the playful game-style background increases engagement. The use of spacing, bold text, and a prominent action button guides the user naturally to the next step. Overall, this screen ensures users understand the task before starting, which improves task performance.

FRAME 2: Chunking Phase (Observation Phase)

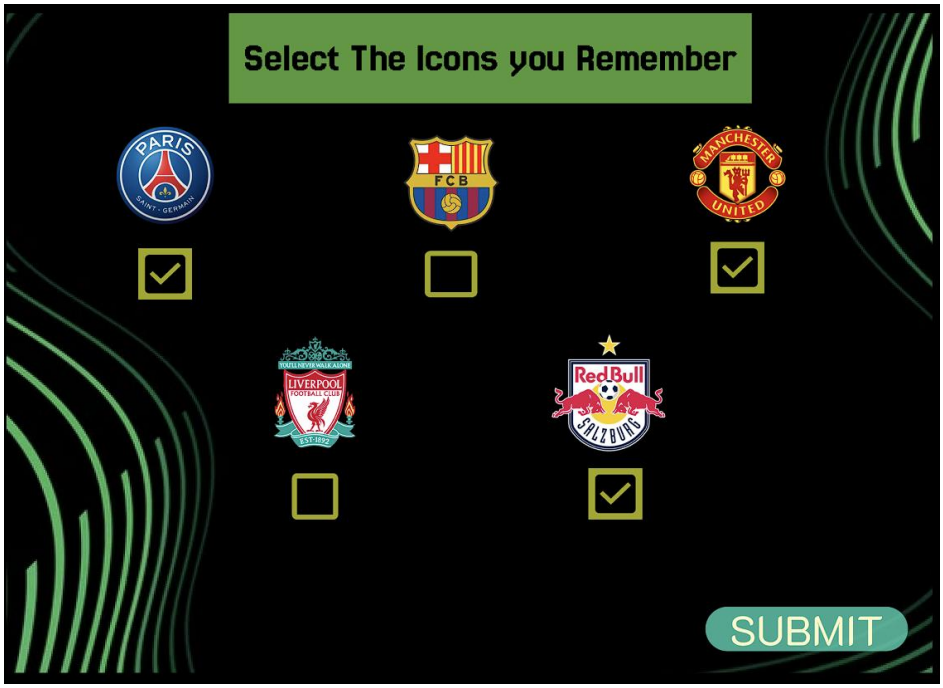
This screen represents the visual encoding stage of the memory recall task. Users are given a limited amount of time to observe multiple icons displayed in a structured grid.

The grid layout plays a key role in chunking. By arranging icons evenly, users can group items mentally based on similarities such as characters, colors, facial expressions, or roles. For example, users may group warriors together, magical characters together, or icons with similar color backgrounds.

The countdown timer creates urgency and encourages focused attention. Time pressure forces users to adopt efficient memory strategies rather than relying on slow repetition. The consistent visual style and spacing prevent distraction, allowing users to concentrate solely on memorization.

This phase demonstrates how chunking transforms a large amount of visual information into manageable memory units.

FRAME 3: Recall Phase (Selection Phase)



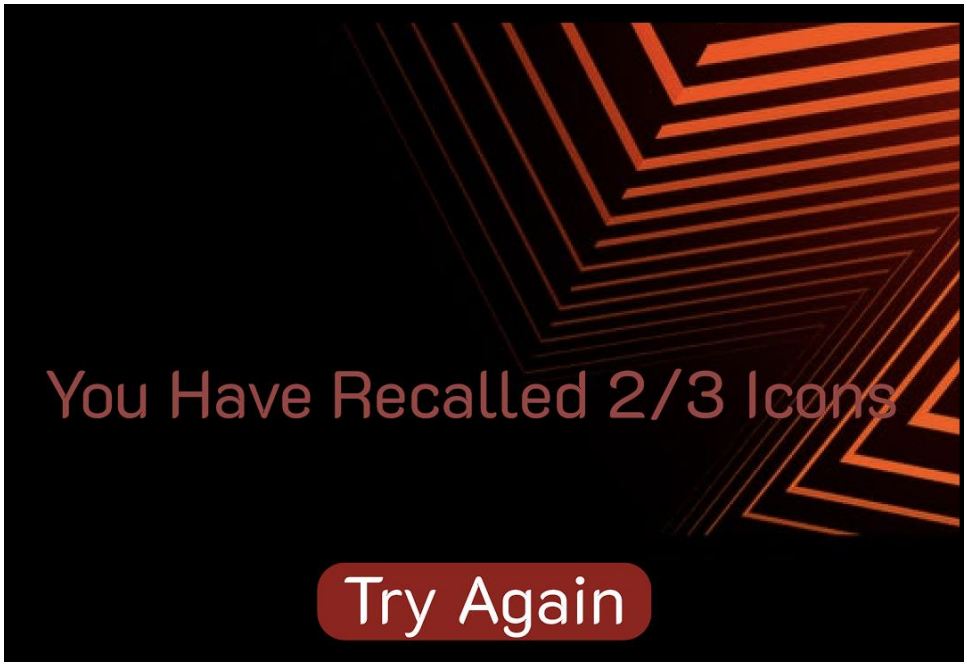
The recall phase tests the user's ability to retrieve stored visual information from short-term memory.

Users are shown a new set of icons that includes both previously seen icons and new distractor icons. This design checks not only recognition but also memory accuracy. Selecting incorrect icons indicates false memory or confusion.

The checkbox-based interaction allows users to carefully evaluate each option. The layout is clean and uncluttered, preventing visual overload during recall. This phase directly measures the effectiveness of the chunking strategy used in the observation phase.

By introducing distractors, the interface simulates real-world memory challenges where users must differentiate between familiar and unfamiliar information.

FRAME 4: Result Page (Feedback Phase)



This screen provides immediate feedback on the user's performance. Showing the score (e.g., 2/3) helps users understand how accurately they recalled the icons.

Instant feedback reinforces learning by allowing users to reflect on their memory strategy. The star-based rating and positive message increase motivation and reduce stress.

Action buttons such as try again or continue give users control over their learning process. Gamified visuals keep the experience engaging while maintaining clarity. This phase completes the memory recall cycle by linking effort with outcome.

Conclusion

This experiment demonstrates how chunking improves visual memory recall by organizing information into meaningful groups. The step-by-step UI design supports attention, encoding, retrieval, and feedback. Such designs are highly effective in educational and cognitive training applications.

PROTOTYPE LINK:

https://www.figma.com/proto/yk67FFJ0fUaQaFv2i7KhpI/Untitled?node-id=1-2&p=f&t=4d1g2ZUQa9y2U8RP-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=5%3A22&show-proto-sidebar=1